

QFT – Lecture 1

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I. Motivation

The goal is to compute decay rates and the probabilities of inelastic processes for any allowed final state.

inelastic collision $\implies$ for any final states.

Context in QM

- The Dirac equation (spin- $\frac{1}{2}$).
- The Klein–Gordon equation (spin-0).

In a single-particle interpretation, these equations describe a particle and its antiparticle. Processes with changing particle number require a field-theoretic description.

SOLUTION QFT is a quantum relativistic theory of fields.

Particles correspond to excitations of modes of these fields.

Features

Relativity

$$v \leq c$$

Causality.

Quantum theory

$$\text{probability} \propto |\text{probability amplitude}|^2.$$

probability amplitude

$$= \sum_{\text{channels}} \text{probability amplitude for the channel.}$$

$$\star \quad \sum \text{probability} = 1.$$

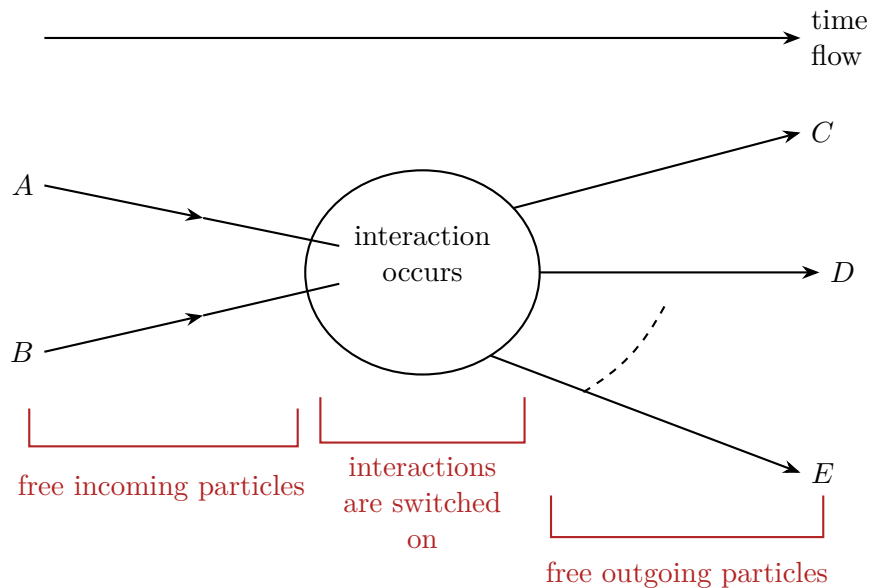
II. First encounter with the S-matrix

New goal: to compute probability amplitudes.

Example: a collision with two incoming particles

$$A + B \longrightarrow C + D + E + \dots$$

Scattering process



Quantum description

The incoming two-particle system is described by a normalized state

$$|\text{in}\rangle, \quad \langle \text{in} | \text{in} \rangle = 1.$$

The interaction takes place, and the scattering operator maps the incoming state to

$$S|\text{in}\rangle.$$

The state $S|\text{in}\rangle$ contains the information of all possible outgoing states.

The scattering operator S :

- Encodes the interaction.
- Acts as an evolution operator on $|\text{in}\rangle$.

Probability amplitude and probability

- The probability amplitude to get the “ C and D and E ” system (the $|\text{out}\rangle$ state) is

$$\langle \text{out} | S | \text{in} \rangle.$$

- Probability of this final state:

$$|\langle \text{out} | S | \text{in} \rangle|^2.$$

Conservation of probability and unitarity

Assume that the outgoing states form a complete orthonormal set. They are linearly independent and satisfy the completeness relation

$$\sum_{\text{out}} |\text{out}\rangle \langle \text{out}| = \mathbf{1}.$$

Conservation of total probability gives

$$\sum_{\text{out}} |\langle \text{out} | S | \text{in} \rangle|^2 = 1.$$

The left-hand side is

$$\begin{aligned} \sum_{\text{out}} \underbrace{\langle \text{in} | S^\dagger | \text{out} \rangle}_{\langle \text{out} | S | \text{in} \rangle^*} \langle \text{out} | S | \text{in} \rangle. \\ \Downarrow \\ \langle \text{in} | S^\dagger \left(\underbrace{\sum_{\text{out}} |\text{out}\rangle \langle \text{out}|}_{\mathbf{1}} \right) S | \text{in} \rangle. \end{aligned}$$

Hence

$$\langle \text{in} | S^\dagger S | \text{in} \rangle = 1.$$

Since this holds for every normalized incoming state,

$$\boxed{S^\dagger S = \mathbf{1}} \quad \text{Unitarity.}$$

III. Some formal reminders

1. A few mathematical formulae

a) The Volterra equation – T-product

From a differential equation to an integral equation

$$\dot{F}(t) = f(t) \implies F(t) = F(t_0) + \int_{t_0}^t f(\tau) d\tau.$$

Differentiating the integral expression and evaluating it at $t = t_0$ verifies the equation and its initial condition.

Example: the Schrödinger equation

$$i\hbar\dot{\psi}(t) = V(t)\psi(t) \iff \underbrace{\dot{\psi}(t)}_{\dot{F}(t)} = \underbrace{\frac{1}{i\hbar}V(t)\psi(t)}_{f(t)}.$$

Applying this identity gives an integral equation for the solution of the Schrödinger equation:

$$\begin{aligned} \psi(t) &= \psi(t_0) + \int_{t_0}^t \frac{1}{i\hbar} V(t_1) \psi(t_1) dt_1 \\ &= \psi(t_0) + \int_{t_0}^t \frac{1}{i\hbar} V(t_1) \underbrace{\left\{ \psi(t_0) + \int_{t_0}^{t_1} \frac{1}{i\hbar} V(t_2) \psi(t_2) dt_2 \right\}}_{\psi(t_1)} dt_1. \end{aligned}$$

Repeating the substitution indefinitely gives the formal series

$$\psi(t) = \psi(t_0) + \sum_{n=1}^{\infty} \left(\frac{1}{i\hbar} \right)^n \int_{(n)} V(t_1) V(t_2) \cdots V(t_n) \psi(t_0),$$

where

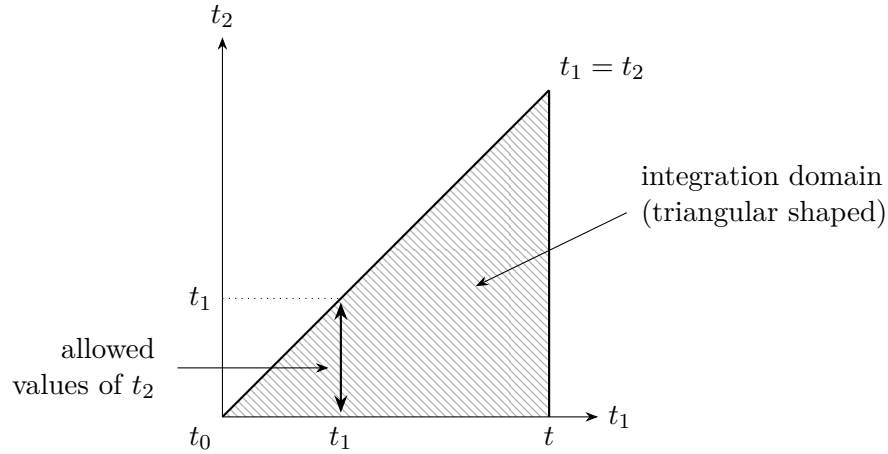
$$\int_{(n)} \equiv \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n.$$

The second-order term

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 V(t_1) V(t_2).$$

This implies that

$$t_0 \leq t_2 \leq t_1.$$



Integration domain of J_2 ; allowed values of t_2 .

We want the boundaries of the integrals not to depend on the integration variables.

Answer: the Heaviside function,

$$\Theta(t) = \begin{cases} 1, & \text{when } t \geq 0, \\ 0, & \text{when } t < 0. \end{cases}$$

Then

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 V(t_1) V(t_2) \Theta(t_1 - t_2).$$

Symmetrization and the time-ordered product

A few extra manipulations give

$$\begin{aligned} J_2 &= \frac{1}{2}J_2 + \frac{1}{2}J_2 \\ &= \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \frac{1}{2} V(t_1)V(t_2)\Theta(t_1 - t_2) + \frac{1}{2}J_2. \end{aligned}$$

In the second term, interchange the dummy variables: $t_1 \rightarrow t_2$ and $t_2 \rightarrow t_1$.

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \frac{1}{2} \left(V(t_1)V(t_2)\Theta(t_1 - t_2) + V(t_2)V(t_1)\Theta(t_2 - t_1) \right).$$

We can now introduce the **T-product (time-ordered product)**.

For two operators:

$$\underbrace{\mathbf{T}}_{\text{time}}[V(t_1)V(t_2)] = \begin{cases} V(t_1)V(t_2), & \text{when } t_1 \geq t_2, \\ V(t_2)V(t_1), & \text{when } t_2 > t_1. \end{cases}$$

Equivalently, for distinct times, the Heaviside function gives

$$\begin{aligned} \mathbf{T}[V(t_1)V(t_2)] &= V(t_1)V(t_2)\Theta(t_1 - t_2) \\ &\quad + V(t_2)V(t_1)\Theta(t_2 - t_1). \end{aligned}$$

The value on the diagonal $t_1 = t_2$ does not affect these ordinary integrals for regular integrands. Hence,

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \frac{1}{2} \mathbf{T}[V(t_1)V(t_2)].$$

The general n th-order term

$$J_n = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \cdots \int_{t_0}^t dt_n \frac{1}{n!} T[V(t_1)V(t_2)\cdots V(t_n)].$$

The solution to the Schrödinger equation

$$\begin{aligned} \psi(t) &= \psi(t_0) + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{1}{i\hbar} \right)^n \underbrace{\int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n T[V(t_1)\cdots V(t_n)]}_{n \text{ time integrations}} \psi(t_0). \\ \implies \psi(t) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{i\hbar} \right)^n \left\{ \int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n T[V(t_1)\cdots V(t_n)] \right\} \psi(t_0). \end{aligned}$$

For $n = 0$, the empty integral and operator product are the identity. The series therefore has the same structure as the exponential series:

$$n = 0 : \quad 1, \quad \sum_{n=0}^{\infty} \frac{1}{n!} x^n.$$

Since this looks like the expansion of the exponential function, we define:

Time-ordered exponential

$$\begin{aligned} &T \exp \left(\int_{t_0}^t f(\tau) d\tau \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n T[f(t_1)\cdots f(t_n)]. \end{aligned}$$

Hence, we can write the solution of the Schrödinger equation as:

$$\begin{aligned} &i\hbar \dot{\psi}(t) = V(t)\psi(t) \\ \iff \psi(t) &= T \exp \left(\frac{1}{i\hbar} \int_{t_0}^t V(\tau) d\tau \right) \psi(t_0). \end{aligned}$$

Remark: If V does not depend on time, then

$$\psi(t) = \underbrace{\exp \left(\frac{1}{i\hbar} (t - t_0) V \right)}_{\text{evolution operator in QM}} \psi(t_0).$$

b) Green functions

Green functions provide a method for solving linear differential equations.

We want to solve

$$Ly(x) = f(x),$$

where L is a linear differential operator

$$\left(\text{e.g. } L = \frac{d^2}{dx^2} \right).$$

If we can solve

$$LG(x) = \underbrace{\delta(x)}_{\text{Dirac delta distribution}}, \quad (*)$$

For a translation-invariant operator, a particular solution is

$$y(x) = \int_{-\infty}^{+\infty} G(x - x') f(x') dx' = \underbrace{(G * f)(x)}_{\text{convolution product}}.$$

The problem is to solve (*) (method by Fourier transform):

$$\text{FT}[\delta(x)] = 1.$$

G is the Green function. Boundary conditions determine the appropriate Green function; a homogeneous solution may be added as required.

2. Canonical transformations in QM

a) Definition

A canonical transformation does not change the physical content of the theory and keeps the structure of the mathematical expression intact. It simplifies the calculation.

In QM, the canonical transformations will be unitary operators U .

States

$$|\psi\rangle \xrightarrow{U} |\psi'\rangle = |U\psi\rangle = U|\psi\rangle.$$

Remark:

$$\begin{aligned}\langle\chi'|\psi'\rangle &= \langle\chi|\underbrace{U^\dagger U}_1|\psi\rangle \\ &= \langle\chi|\psi\rangle.\end{aligned}$$

The corresponding bra is the Hermitian adjoint of the ket:

$$\langle\chi'| = |\chi'\rangle^\dagger.$$

Observable (operator)

$$A \xrightarrow{U} A' = UAU^\dagger.$$

b) Reminder: Schrödinger picture

State

 $|\psi(t)\rangle$

$$\begin{cases} i\hbar \frac{\partial |\psi\rangle}{\partial t} = H|\psi\rangle, \\ |\psi(t_0)\rangle = |\psi_0\rangle. \end{cases}$$